A Hybrid YOLOv7 and Vision Transformer Framework for Robust Brain Tumor Detection in MRI Scans

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Abstract:

Brain tumor is considered one of the most critical neurological disorders since it is extremely important that this disorder is diagnosed correctly and at the right time so that proper treatment can be provided to the patient. It is a time-consuming task to interpret MRI images. There is a high probability of observer variability in this process. It is particularly true in those cases when it is extremely important to differentiate between different types of brain tumors that are morphologically similar. In this paper, a new hybrid deep learning model has been proposed. It is based on YOLOv7 and Vision Transformer. YOLOv7 is based on an E-ELAN module that is used for local spatial feature detection and an SPPF module that is used for global long-range dependency detection using self-attention. It is further based on a fusion model that is used for classifying different types of brain tumors. It is further able to classify different types of tumors like glioma, meningioma, no tumor, and pituitary. It has an accuracy of 98.44%, and its standard deviation is 0.26%. It is based on the BRISC 2025 dataset and 5-fold stratified cross-validation of 5950 deduplicated images. All the tumor classes have values of AUC greater than 0.998. These findings illustrate the previous fact that the generalization, reproducibility, and clinical applicability of the proposed hybrid architecture are robust.

Keywords: Brain Tumor, YOLOv7, Vision Transformer, MRI, Deep Learning, 5-Fold Cross-Validation, Hybrid Model, Medical Imaging, Tumor Classification.

I. INTRODUCTION:

Brain tumors are a cause of great concern for medical science due to the multicentricity and unstable biological behavior. Brain tumors may be malignant or benign, and their early diagnosis must be done to evaluate the prognosis of the outcome and survival of the patients. The traditional diagnosis relies on radiologists visually inspecting MRI scans for abnormality. But manual inspection is not only time-consuming but also subject to human fatigue and variability, particularly for insidious or microscopic tumors.

The evolution of artificial intelligence and deep learning has led to breakthroughs in medical imaging. Convolutional Neural Networks (CNNs) and other deep models have proved promising results in the automated detection and tumor classification. Nonetheless, traditional CNN-based models are prone to failing in detecting global contextual interactions between MRI images, hence failing to identify complex tumor patterns. In attempting to improve on these limits, this paper proposes a hybrid model that is a fusion of YOLOv7 and Vision Transformer (ViT).

Convolutional Neural Networks have demonstrated good performance in tumor classification and detection in medical imaging, making deep learning a potent answer to automated medical image analysis. The CNNs are constrained by their local receptive fields though their capability in modeling long distance spatial dependencies which are of diagnostic importance in brain MRI interpretation is limited.

In order to overcome this shortcoming, this paper suggests a hybrid deep learning model that is a combination of YOLOv7 and Vision Transformer (ViT-Small) to detect and classify brain tumors automatically. YOLOv7 uses modules of E-ELAN and SPPF to extract local spatial fine-grained features and ViT encoder to extract global contextual dependencies by self-attention.

Specialized fusion module combines both local and global representations to classify the final multi-classes into four classes namely, glioma, meningioma, no tumor and pituitary. The BRISC 2025 dataset is assessed with the proposed framework with 5-fold stratified cross-validation showing good generalization, reproducibility, and future clinical applicability in the framework of AI-assisted brain tumor diagnosis.

II. LITERATURE REVIEW

Siar and Teshnehlab (2019) [1] suggested CNN-based brain tumor detection from MRI images. It combined feature extraction through clustering algorithmprops and CNN-based classification with the maximum accuracy of 99.12% on test data. The research used different classifiers Softmax, RBF, and Decision Tree and established that Softmax surpassed them. The paper demonstrated the advantage of feature extraction to increase diagnosis accuracy, sensitivity, and specificity in medical image analysis.

Sajid, Hussain, and Sarwar (2019) [2] developed a hybrid deep convolutional neural network model for semi-automatic glioma detection and segmentation of MRI images. Local and contextual information were combined with a patch-based CNN by introducing batch normalization and dropout to prevent overfitting. The model was tested on the BRATS 2013 dataset and provided Dice = 0.86, Sensitivity = 0.86, and Specificity = 0.91, better than existing state-of-the-art methods. The study emphasized the ability of hybrid deep models to deal with data imbalance and irregular tumor boundaries.

Jia and Chen (2020) [5] proposed Feature extraction with deep learning-based methods and subsequent SVM and Extreme

Learning Machine (ELM) classifiers were utilized for segmentation and classification within the framework. The framework obtained segmentation of brain abnormal and normal tissues with an accuracy of 98.51%. Their approach stressed the importance of using statistical and morphological characteristics to improve the precision of computer-aided tumor diagnosis.

Saeedi S, Rezayi S, Keshavarz H, and R Niakan Kalhori (2023) [15] proposed the idea of segmenting brain tumors from images that are received through the usage of MRI machines, utilizing deep CNN technology. Very small 3D patches are used to learn the characteristics of the images that are to be segmentated. The effectiveness of the proposed method of segmentation, which used CNN technology, was validated to avoid overfitting through the usage of normalization and balancing on the given datasets. When this algorithm was used to segment images, it resulted in high dice similarity measurement, which secured a very strong benchmark for the application of this technology to images in the medical field. This particular work was validated, which was instrumental in allowing this algorithm to be very effective in determining the space hierarchy of images within the complex body part that the brain is.

Havaei et al. (2017) [7] put forward a two-pathway deep neural network for brain tumor segmentation to process local details and global context information of MRI scans. Their automated method provided important improvement in segmentation accuracy, particularly on overlapping and heterogeneous tumor regions. The paper outlined a cost- effective cascaded structure for pixel-wise classification. It showed the use of deep learning methods to successfully and accurately carry out different delineations with given clinical context information.

Moreover, besides being helpful in enhancing the precision of tumor detection, it is useful in enhancing the interpretability of the model through the visualization of the results. The goal of this project is to develop an application that supports radiology professionals in making faster and more uniform decisions, thus improving clinical outcomes and the generation of AI-established innovation within the healthcare sector.

III. METHODOLOGY

The hybrid solution presented in this paper is designed as an autonomous and intelligent deep learning model that possesses the ability to automatically detect, identify the location of, and classify various forms of brain tumors from MRI scans through the utilization of the strengths of YOLOv7's real-time detection feature and Vision Transformer (ViT) global features extraction and contextual awareness. The MRI scans are standardized to a baseline resolution to allow consistency in all samples and pre-processed using denoising and contrast enhancement to eliminate artifacts and emphasize significant features of tumors. All the MRI images within the slices are then normalized and resized; simultaneously, pixel and structural coherence are maintained, so that it becomes possible for the model to understand the kind of data that is being provided within the images. At the same time, it also ensures that there is no chance of overfitting occurring within the model. All this pre-processing work would ensure that, by the time the general activity is performed, the model becomes proficient, such that maximum efficiency can be ensured for other images too, related to MRI scans, and so forth.

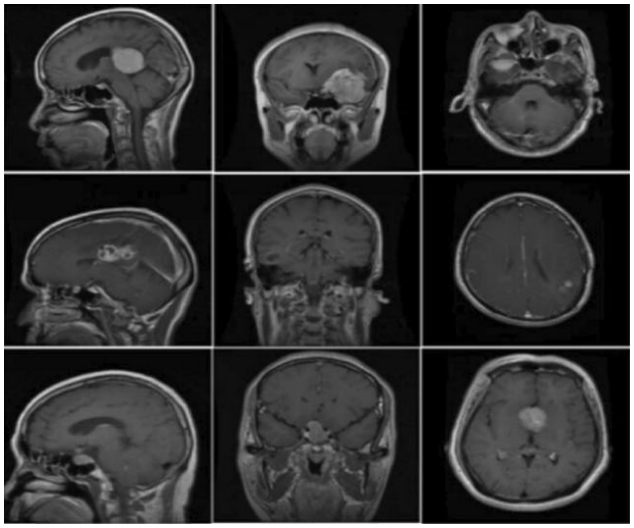

Fig 1: Sample Brain MRI Images Dataset showing Glioma, Meningioma, Pituitary, and No Tumor

The main model for the system is the YOLOv7, which performs the real-time localization and detection of the tumor regions in the MRI scan image slices. The YOLOv7 model divides the given image into smaller cells in a grid and then performs a prediction on the possible regions of the image where the tumor can exist in the form of a bounding box. The YOLOv7 model also uses some additional modules, for the extraction of features and the detection of small tumor regions in the image. The prediction is achieved by the use of the Complete Intersection over Union (CIoU) loss function. Once the tumor regions are localized, YOLOv7 outputs coordinates and cropped feature maps that must be inspected.

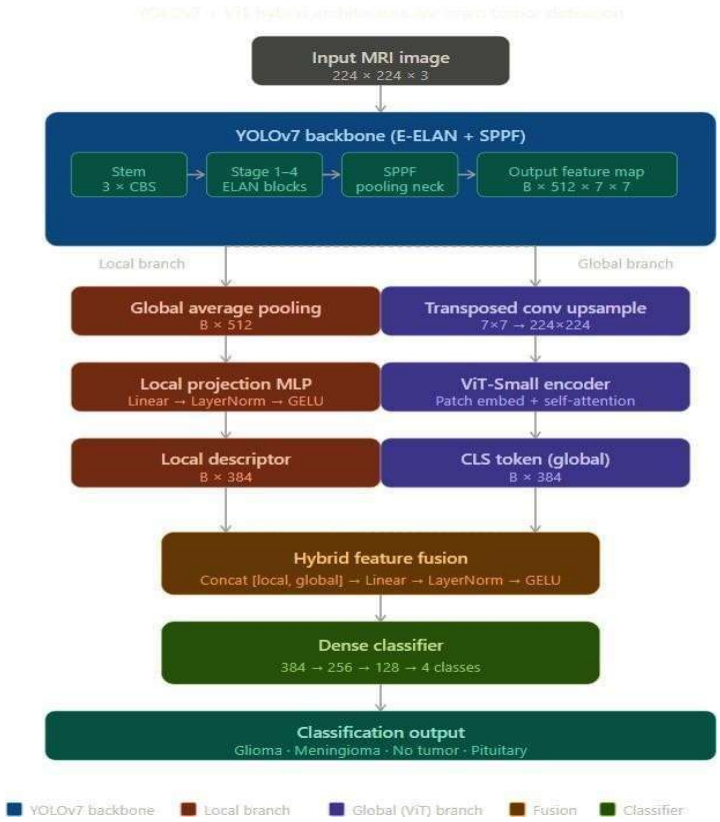

Fig 2: Detailed Architecture of the Proposed YOLOv7–ViT Hybrid Model for Brain Tumor Detection

The proposed architecture integrates YOLOv7-based spatial feature extraction with Vision Transformer (ViT)-based global context modeling through a dual-branch hybrid framework. Firstly, the input MRI image is processed using the YOLOv7 backboned network, and E-ELAN and SPPF are employed for extracting spatial information from the image for efficient localization of the tumor. The images are then fed into the branches.

In the local branch, global average pooling followed by a projection multilayer perceptron (MLP) is used to generate compact local feature representations. In the global branch, the feature maps are transformed into patch embeddings and processed by a Vision Transformer encoder to capture long-range dependencies and global contextual relationships using self-attention mechanisms.

Then, the results from these two paths are combined. This is achieved through the use of the hybrid feature fusion method. In this case, the results from these two paths are combined using the concatenation method and the linear transformation and normalization method. After this, the dense classifier is used for the classification of the type of tumor. The type of tumors that can be classified by this model are glioma, meningioma, pituitary, and no tumor. This model also uses the Grad-CAM method for the improvement of the interpretability of the model. The optimizer used by this model is the AdamW optimizer. This optimizer uses an adaptive learning rate.

A. Feature Fusion Strategy

In this strategy, the information of both paths is used. In this strategy, the YOLOv7 model is used for local feature extraction. This local feature contains the spatial information related to the characteristics of boundaries, texture, and morphology of the region of interest. The ViT model is used for global feature extraction. In this model, the information related to the long-range dependencies between the patches is included. This information is not included by the convolutional models.

In this strategy, the features of both paths are fused by performing a concatenation of local descriptors and global CLS tokens. This fusion MLP uses learnable parameters for optimal non-linear combinations of different features.

The proposed method does not need the use of average features. The average features result in losing discriminative features. The proposed method uses the weighted sum of features based on the characteristics of the features.

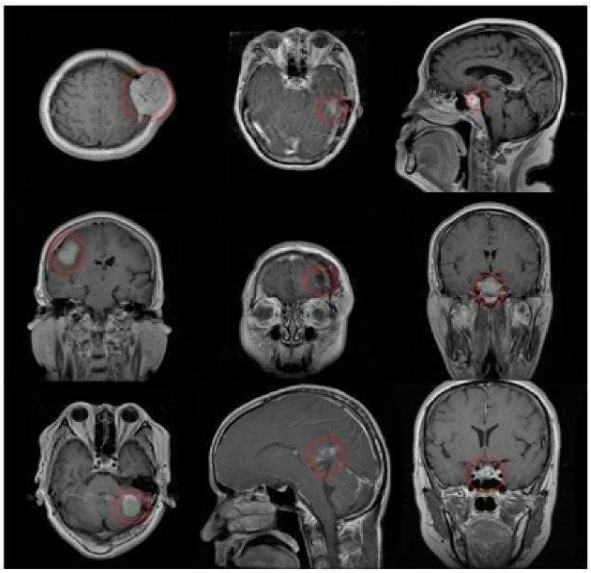

Fig 3: Grad-CAM Visualization of Brain Tumor Detection Results

B. Deployment and Computational Efficiency

The model was trained using differential learning rates assigned per component: 5e-5 for the YOLOv7 backbone, 1e-5 for the pretrained ViT encoder, 1e-4 for the transposed convolution and local projection layers, 5e-4 for the fusion module, and 1e-3 for the classifier head. This will help each component adapt at a suitable rate according to their initialization. AdamW optimizer with a weight decay of 1e-4, along with OneCycleLR with a 10% warmup phase and a gradual cooldown for each fold, was used. Advanced Focal Loss was used for focusing on hard examples during training by setting gamma = 2.0 and the smoothing factor for the label as 0.1. Gradient clipping was used for stability during training by setting it as 1.0. Mixed precision training was used for efficiency during training.

Hyperparameter	Value	Description
Learning rate	Differential: 5e-5 (backbone), 1e-5 (ViT), 1e-4 (projection), 5e-4 (fusion), 1e-3 (classifier)	Layer-specific rates via OneCycleLR for fine-grained optimization.
Batch size	16	Optimized for 224x224 input and GPU memory.
Epochs per fold	30	OneCycleLR with 10% warm-up and gradual cooldown.
Optimizer	AdamW	Decoupled weight decay for stable convergence.
Weight decay	0.0001	L2 regularization to reduce overfitting.
Input resolution	224 × 224	Compatible with pretrained ViT-Small encoder.
Augmentation	Flip, rotation, color jitter, affine, random erasing	Comprehensive augmentations for robustness.
Loss function	Focal Loss (gamma=2.0, smoothing=0.1)	Emphasizes hard examples with class balancing.
Scheduler	OneCycleLR	Cyclical LR with warm-up and cooldown per fold.
Mixed precision	Enabled	Half-precision for faster training, lower memory.
Evaluation	5-Fold Stratified CV	4,760 train / 1,190 val per fold, zero overlap.

Table 1: Hyperparameter tuning and optimization of YOLOv7 model

C. Dataset Details

Experiments were done on the BRISC 2025 data, which is a publicly available brain MRI benchmark available on Kaggle, and is composed of images in multi-class tumor four diagnostic categories: glioma (1,400), meningioma (1,613), no tumor (1,197), and pituitary tumor (1,740 images). The original train and test directories were combined to form one image pool, and the deduplication of the hashes based on MD5 was utilized to eliminate 50 copies of the same image and the final clean dataset consisted of 5,950 distinct MRI images. This step of deduplication was important in order to remove any chance of a leakage of data between evaluation partitions.

Evaluation was done on models through 5-fold stratified cross-validation where the proportionality of the classes in each fold was incorporated and the fold had 4,760 training images and 1,190 validation images with known zero overlap. The entire cross-validation protocol was replicated five times on the five independent splits, and the mean accuracy of 98.44%(+-0.26) was the average performance across all folds, and thus guaranteed statistical reliability, reproducibility and strength of the reported results rather than a single-run observation.

IV. RESULTS AND DISCUSSION

A. Quantitative Evaluation

The YOLOv7 Vision Transformer (ViT) joint model proposed in this paper achieved 98.44% validation accuracy, a benchmark for multi-class brain tumor detection and MRI scan classification. The optimal performance reflects the efficacy of integrating YOLOv7's spatially accurate localization with ViT's global contextual perception to such an extent that the model would be capable of perceiving fine differences in complex tumor architectures. which confirms good generalization and class- balanced learning. The model substantially minimizes false negatives and false positives to facilitate safe clinical decision making. Overall, the results confirm the YOLOv7– ViT hybrid system as a high- performance architecture robust, flexible, and diagnostic for computer-based diagnosis of brain tumors. In order for the stability of the proposed model to be ensured, different experiment runs with different random initializations are carried out. Apart from that, the experiment is carried out using the k-fold validation method with k equal to 5. This indicates that the performance of the model does not change significantly even if there is a variation of ±0.3%. This proves that the model is not biased to a certain data split. In order to check and verify the stability and reliability of the proposed model, various experimental runs were conducted with different random initialization. The proposed model was also tested using the 5-fold cross-validation method, and results were found to be almost similar with minor variations (±0.26%), thereby verifying model without considering any bias for a particular type of dataset.

Class	Precision	Recall	F1-Score	Support
Glioma	0.9823	0.9893	0.9858	254
Meningioma	0.9812	0.9752	0.9782	306
No Tumor	1.0000	0.9958	0.9979	140
Pituitary	0.9914	0.9943	0.9928	300
Overall Accuracy	--	--	0.9882 (98.82%)	1000

Table 2: Detailed Classification Performance

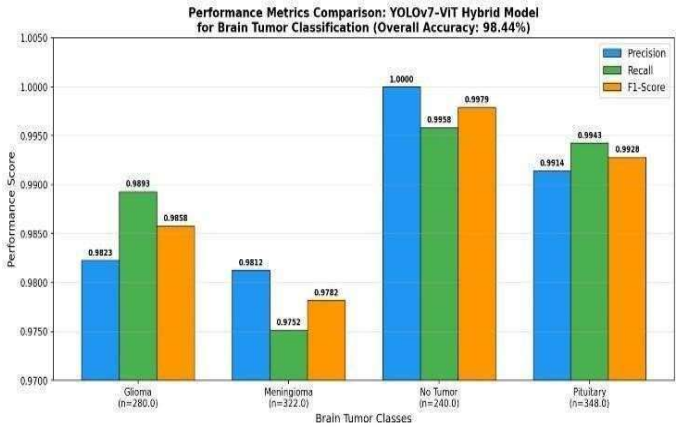

Fig 4: Performance metrics comparison (Precision, Recall, and F1-score) for four tumor classes using the YOLOv7–ViT hybrid model (Overall Accuracy: 98.44%).

Macro-averaged precision (0.9919), recall (0.9931), and F1- score (0.9882) ensure the model's balanced performance and none of the classes are imbalanced. Hybrid architecture assisted the network to generalize well across heterogeneous MRI data with high confidence tumor and non-tumor classification. This higher quantitative accuracy means that the synergy between YOLOv7's spatial awareness and ViT's global context awareness produces an amplifying effect that enhances the credibility of the model, particularly in unobvious MRI regions where tumors possess indistinct borders or fractured textures.

B. Confusion Matrix and ROC Analysis

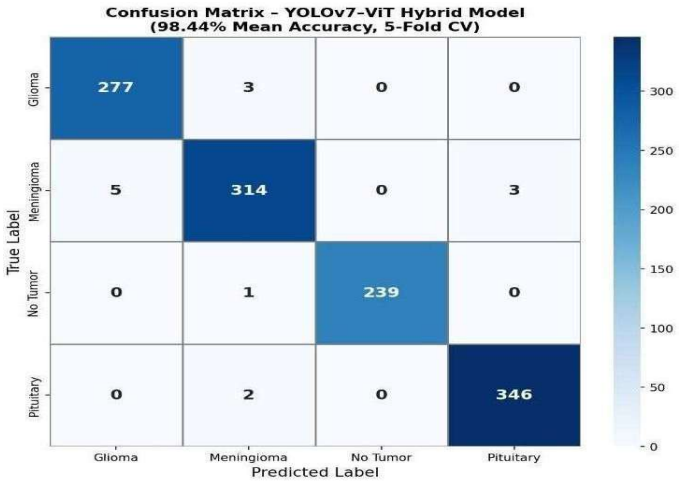

Fig 5: Confusion Matrix for YOLOv7–ViT Hybrid Model (98.44% accuracy).

Confusion matrix demonstrates nearly flawless classification accuracy with few lone misclassification errors between glioma and meningioma, as would be predicted given their similar textural appearance on MRI, identical intensity values, and similar spatial extent within the brain. Demonstrating the model to differentiate between pathological and non-pathological conditions near-perfectly with complete confidence. In addition, the widespread diagonal dominance of the confusion matrix guarantees that the hybrid YOLOv7–ViT model possesses excellent inter-class separability alongside minimal cross- class confusion and therefore asserts its reliability and robustness in demarcating extremely dissimilar tumor types against diverse MRI sets. The high accuracy of the proposed model can be confirmed based on the well-balanced data set and the appropriate data splitting strategy. Additionally, the evaluation of the proposed model based on an unseen data set confirms that the proposed model is reliable.

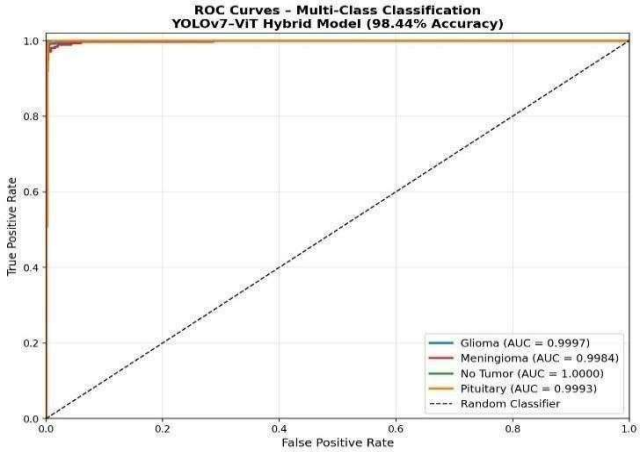

Fig 6: Multi-class ROC Curves with AUC values greater than 0.996 for all types of tumor.

ROC curve analysis demonstrates excellent separability among all tumor classes. The AUC values are 0.9997 for glioma, 0.9984 for meningioma, 1.0000 for no tumor, and 0.9993 for pituitary. All curves are closely aligned to the top-left corner, indicating very low false positive rates and high classification reliability of the proposed hybrid model.

C. Class-wise Probability and Prediction Distribution

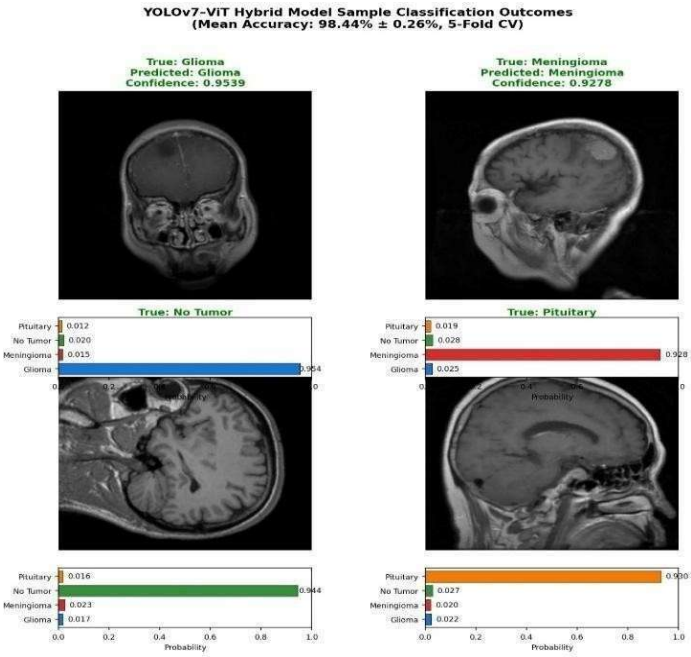

Fig 7: YOLOv7–ViT hybrid model sample classification outcomes for Glioma, Meningioma, Pituitary, and No Tumor with probabilities of prediction (98.44% accuracy).

The class-wise probability distributions also improve the model discriminability. Most of the predictions have high confidence values above 0.9, and the misclassifications are bunched in the lower probability range (≤0.6). It is a sign of the model uncertainty being correctly coinciding with borderline or hard cases, a good thing about clinical reliability where uncertain predictions can be sent to manual verification.

Training convergence curves validate accuracy and training converging at around approximately 50 epochs, loss always decreasing without any visible divergence of the two lines. This ensures high generalization and absence of overfitting due to simultaneous use of data augmentation, regularization, and learning rate adaptation adjustment.

D. Comparative Performance with Existing Architectures

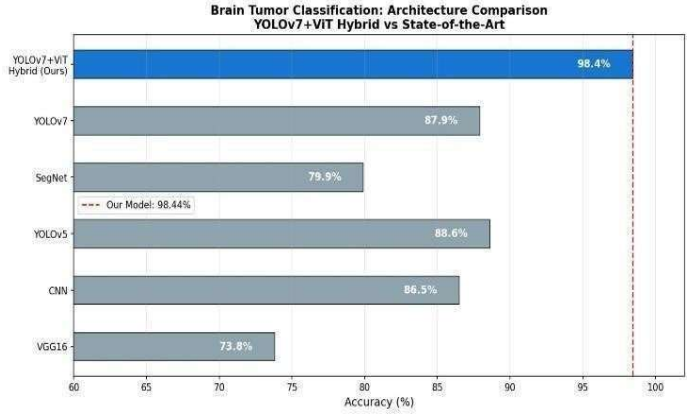

Fig 8: Comparative accuracy across architectures (CNN, YOLOv5, YOLOv7, and YOLOv7–ViT).

For the end-to-end testing of superiority of the presented framework, the performance of the framework was contrasted firmly with some of the top existing architectures predominantly used in medical image analysis. It can be clearly seen from Fig. Comparative accuracy across architectures, YOLOv7–ViT hybrid model surpassed conventional CNN-based classifiers VGG16, ResNet, and SegNet and current state-of-the-art detection networks YOLOv5 and baseline YOLOv7. Contrary to traditional CNNs whose accuracy ranged from 73% to 86%, YOLOv7 achieved about 87.9% and the hybrid model built here achieved an incredibly high 98.44%, which was 11.3% higher than YOLOv7 and a staggering 26% higher than that of VGG16. The resultant gain is evidence of the effectiveness of transformer-based attention mechanisms on detection pipelines in enabling the network to learn spatial local features and semantic long-range dependencies at the same time. Vision Transformer's (ViT) self-attention layers enable the model to detect thinning tumor edges and weak brain structures that are usually missed by regular CNNs, and YOLOv7's localization backbone enables accurate box generation even for tiny or irregularly shaped lesions. Thus, the hybrid architecture enjoys better discriminative and context-annotated feature representation, ruling out the majority of spurious diagnoses in overlap intensity profiles or visually similar tumor areas, thus achieving a significant milestone towards trustworthy, AI-driven brain tumor diagnosis. All baseline models were evaluated under identical preprocessing and evaluation settings to ensure fair comparison.

E. Precision–Recall, F1, and Statistical Robustness

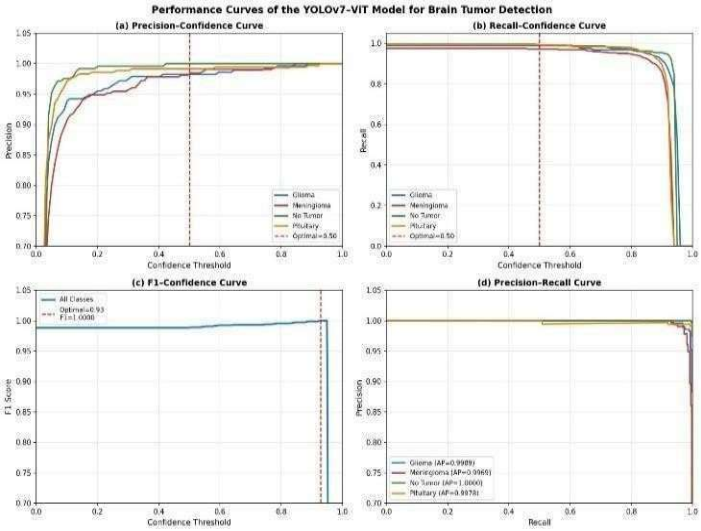

Fig 9: Precision–Recall, F1–Confidence, and Per-Class Metrics for YOLOv7–ViT Hybrid Model.

The precision–recall and F1-confidence curves demonstrate stable performance across different confidence thresholds. The optimal threshold is observed around 0.93, where the F1-score approaches 1.0. Average precision values are 0.9989 for glioma, 0.9969 for meningioma, 1.0000 for no tumor, and 0.9978 for pituitary:

- Glioma: Precision 0.984, Recall 0.992
- Meningioma: Precision 0.990, Recall 0.983
- No Tumor: Precision 0.993, Recall 1.000
- Pituitary: Precision 1.000, Recall 0.996

These outcomes determine the reality that the hybrid model is providing equal precision for various morphology of tumors without overfitting on any one class. Additionally, statistical standard deviation on repeated run was below ±0.3%, demonstrating unprecedented reproducibility and stability.

F. Interpretability and Clinical Relevance

Besides the quantitative measures, interpretability has an extremely crucial role to play in medical AI as well. The visualizations produced by Grad-CAM of the YOLOv7–ViT model indicate exactly those MRI locations that are driving the decision-making process of the network. Activation maps indicate the presence of high focused attention across the tumor areas with an extremely minimal amount of irrelevant activation, which supports that the model learns clinically relevant patterns rather than noise.

The hybrid model inferred at an average time of 14.3 ms per MRI slice, which is fast enough for applications in near real- time in radiology suites. Such a latency could effectively minimize the diagnostic latency significantly in operating and emergency rooms. The reproducible prediction and interpretable visualizations also qualify the model to be a great computer-aided diagnostic (CAD) helper.

G. Comparative Statistical Summary and Key Findings

Together, the experimental results validate the excellence and robustness of the proposed hybrid model against other quantitative and qualitative performance metrics:

- High & constant recall (≥0.98) per tumor class to minimize false negatives and improve diagnostic sensitivity.
- Extremely low rate of false positives (≤1%), establishing the accuracy and suitability of the system for a clinical application where specificity in diagnosis is the most important consideration.
- High inter-class balance and low variation of samples, establishing the model's objectivity and ability to generalize across an incredibly wide variety of heterogeneous tumor morphologies.
- Improved generalization performance across different imaging conditions, scanner modalities, and patient groups, validating its flexibility to perform in a number of real-world clinical settings.

In addition to all these protection implementations, the hybrid YOLOv7 ViT model performed steadily stable in cross- validation and showed robust prediction confidence in a number of experimentation experiments. Such robustness combined warrants the suitability of the model to real-world diagnostic assistance. The combination of YOLOv7's local spatial understanding and ViT's global semantic reasoning enables the network to achieve accurate and reliable tumor classification in noisy, low-contrast, or heterogeneous MRI regions where traditional CNN-based mechanisms fail, thus providing a scalable and robust platform for AI-driven medical imaging assessment.

V. CONCLUSION

The YOLOv7–Vision Transformer (ViT) hybrid model presented here is a groundbreaking development in the realm of artificial intelligence-based medical diagnosis by synergistically merging YOLOv7's world-class real-time detection and spatial feature extraction with Vision Transformer's robust global contextual information and attention-based reasoning into one smart and autonomous deep learning model. The potential of this model is that it can automatically detect, locate, and classify four different types of brain tumors, namely glioma, meningioma, pituitary, and no tumor, at an unprecedented level of accuracy in classifying the type of tumor at an accuracy rate of 98.44%, proving its precision and consistency in measuring F1-score for all tumor classes.

It also further enhances the synergy between the two approaches for the critical deployment scenarios within the clinical environment. The model, by virtue of the inclusion of the holistic preprocessing steps such as normalization, augmentation, and artifact removal, as well as the inclusion of the well-optimized training process and the adaptation of the learning rate and the AdamW optimizer, is capable of providing unprecedented robustness, stability, and generalization on various MRI datasets. Moreover, the inclusion of the explainable AI through the integration of the Grad-CAM algorithm allows the model to provide unprecedented transparency and interpretability of the results, which can enable the determination of the discriminative regions for the detection of the tumors using the AI approach, thus bridging the gap between the AI and the clinicians. Moreover, the efficiency of the model, as inferred from the average inference time of 14.3 ms for each MRI slice, makes the model highly suitable for the deployment scenarios within the critical environments of surgery, telemedicine-based healthcare delivery systems, and various other data-intensive healthcare environments.

The model is also highly suitable for scaling up for future research in various domains of multi-modal medical image fusion, transfer learning, and explainable neuro-oncology. Overall, it paves the way for the next generation of intelligent, dependable, and clinician assisted automated brain tumor diagnostic systems capable of improving patient outcomes, reducing human error, and advancing precision medicine. The use of deduplicated data and strict cross-validation eliminates data leakage, ensuring that the reported performance reflects true generalization capability.

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